

25 Some bond lengths are listed.

bond	H–H	C–H	C–C	C=C	C≡C
bond length/nm	0.074	0.109	0.154	0.134	0.120

Ethyne, C_2H_2 , is a linear molecule. No lone pairs of electrons are present on any of the atoms in the molecule.

What is the distance between the two nuclei of the hydrogen atoms in this molecule?

- A** 0.194 nm **B** 0.338 nm **C** 0.352 nm **D** 0.372 nm